

Creation and Evaluation of Human Models with Varied Walking Ability from Motion Capture for Assistive Device Development

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CONTRIBUTION

Understanding how deep reinforcement learning training parameters affect the ability of the simulated human walking policies to recover from perturbation.

- ◆ Control policies trained with or without perturbation produce gait that resembles that of the human.
- ◆ Varying perturbation parameters influences the walking ability and recovery ability from perturbation.
- ◆ Control policies show adaptability to forces experienced similar to those produced by balance assistive devices.

BACKGROUND

- ◆ As the world ages, rehabilitation and assistive devices will play a key role in improving mobility.
- ◆ Designing controllers for these devices has several challenges, from varying degrees of impairment to unique adaptation strategies of the user.
- ◆ To use computer simulation to address these challenges, simulating human motions is required.
- ◆ Deep reinforcement learning has successfully generated walking motions whose goal is to produce stable human walking policies.
- ◆ However, from a rehab perspective, it is more important to align the walking policy with the impaired person's abilities.

RESULTS

Table 1: Name of Each Control Policy with the Various Perturbation Parameters

Name	Force Range	Start Time	Duration
CP-BASE	0N	Not Applicable	Not Applicable
CP-A	[0N, 50N]	0.1s onwards	Till Termination
CP-B	[0N, 200N]	0.1s onwards	Till Termination

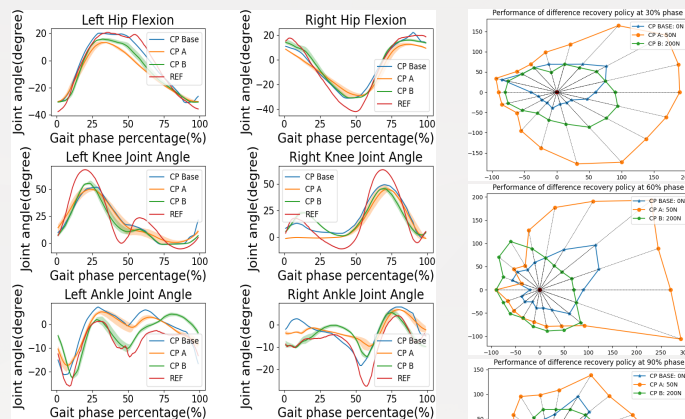


Fig 1: Comparison of mean and standard deviation of lower limb joint angles of CP-BASE, CP-A and CP-B against reference motion along the sagittal plane

Table 2: Comparison of Stride Length between Reference Motion, CP-BASE, CP-A and CP-B

Name	Mean(m)	Min(m)	Max(m)
Reference	1.56	Not Available	Not Available
CP-BASE	1.33	1.27	1.38
CP-A	1.21	1.17	1.35
CP-B	1.10	1.08	1.12

Fig 2: Range of external perturbation required for each control policy before it fails to recover across a range of $[-\pi, \pi]$ at 30% (heel rise), 60% (toe-off) and 90% (tibia vertical) of the gait cycle. Positive X is the forward direction of the model. X and Y axes represent magnitude of force (N).

METHODOLOGY

3D Personalised Human Model

- ◆ Full-body, joint-torque actuated skeletal model with accurate anthropometrical information created from motion capture data and simulated in MuJoCo.
- ◆ Subject is male, at 1.68 meters and weighs 59.9 kg.



Human Walking Policy

- ◆ Model-free deep reinforcement learning approach to develop a human control policy trained on walking motion capture data based on the adversarial motion prior method.¹

Varying Walking Policy's Recovery from Perturbation

- ◆ Perturbation introduced to increase diversity of samples during training, encouraging the model to generalise across different perturbation scenarios.
- ◆ Perturbation can be varied based on position, timing, duration, magnitude and direction of force.

WALKING SEQUENCE

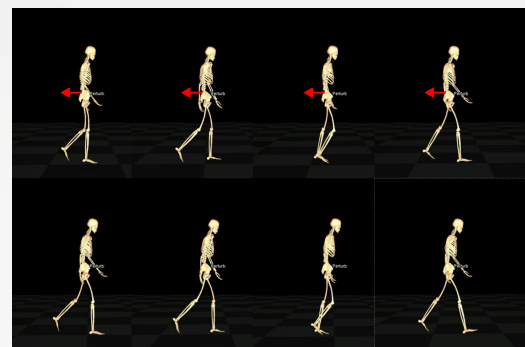


Fig 3: Illustration of change in step length as a result of perturbation. Note that the top has a smaller stride length for all images from left to right
Top: 50N perturbation towards the back
Bottom: No perturbation

DISCUSSION

- ◆ CP-BASE, CP-A and CP-B produced gait patterns similar to reference motion with minor difference in range of joint angles. This difference is due to CPs prioritising stability and balance over exact replication.

- ◆ CP-BASE and CP-A are more stable than CP-B when resisting perturbation. We posit that large forces early on leads to CP-B's inability to learn a nominal gait.
- ◆ When CPs are perturbed rearwards, reduced step length is consistent with behaviour seen when using an overground gait robot.² Conversely, during forward perturbation, step length increases.

ACKNOWLEDGEMENT

- ◆ The project is supported by the Rehabilitation Research Institute of Singapore (RRIS) and A* STAR under its National Robotics Programme (Award: M22BK0074).
- ◆ Participation in the motion capture was voluntary and informed consent was obtained. IRB-2018-04-014 governs the motion capture, and access to data to create the 3D human model is governed by IRB-2022-713, approved by Nanyang Technological University's Institutional Review Board.

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[1] Xue Bin Peng, Ze Ma, Pieter Abbeel, Sergey Levine, and Angjoo Kanazawa. AMP: Adversarial motion priors for stylized physics-based character control. ACM Transactions on Graphics (TOG), 40(4):1–20, 2021.

[2] Lai Li, Ming Jiao, Jiaze Chen, Kuan Yee Tan, Jinyang Cai, Rishir Swaminathan, Karen Sui Geok Chua, Seng Kwee Wee, Christopher Wee Keong Kuah, Huijing Zhou, et al. Mobile robotic balance assistant (mibal): a gait assistive and fall intervention robot for daily living. Journal of NeuroEngineering and Rehabilitation, 20(1):1–17, 2023.